Q 1. Given a string S, for every index i, find the length of longest substring from it which is equal to $\underbrace{\text{prefix}}_{\text{starts from } 0^{th} \text{ index}}$ of whole string.

str ⇒

| 0 | 1 | 2 | 3 |
|---|---|---|---|
| a | b | c | d |

prefix of str ←

- a ✓
- ab ✓
- abc ✓
- abcd ✓

bcd ✗

acd ✗

str ⇒

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 |
| x | x | y | z | x | x | y | z | w | x | x | y | z | x | x | y | z | x |
| ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ |
| 18 | 1 | 0 | 0 | 4 | 1 | 0 | 0 | 0 | 8 | 1 | 0 | 0 | 5 | 1 | 0 | 0 | 1 |

```
for( i=1; i<N; i++){
        j=i
        k=0
    while( j<N && s[j] == s[k]) {
            j++
            k++.
    }
}   z[i] = k
```

$$\boxed{\begin{array}{l} TC \Rightarrow O(N^2) \\ SC \Rightarrow O(N) \end{array}}$$

| | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 |
| | x | x | y | z | x | x | y | z | w | x | x | y | z | x | x | y | z | x |
| | X | 1 | 0 | 0 | 4 | 1 | 0 | 0 | 0 | 8 | 1 | 0 | 0 | 4+1 | 1 | 0 | 0 | 1 |
| | | BF | BF | BF | BF | | | | BF | BF | | | | BF | | | | BF |

| | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | 3 | 4 | 5 | 6 | 7 | |
| | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 |
| | x | x | y | z | x | x | y | z | w | x | x | y | z | x | x | y | z | x |
| | X | 1 | 0 | 0 | 4 | 1 | 0 | 0 | 0 | 8 | 1 | 0 | 0 | 4+1 | 1 | 0 | 0 | 1 |
| | | BF | BF | BF | BF | | | | BF | BF | | | | BF | | | | BF |

| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 |
| Str : | x | x | y | x | x | y | x | x | a | x | x | y | x | x | z |
| | X | 1 | 0 | 5 | 1 | 0 | 2 | 1 | 0 | 5 | 1 | 0 | 2 | 1 | 0 |

i

| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 |
| | x | x | y | x | x | y | x | x | a | x | x | y | x | x | z |
| ans → | X | 1 | 0 | 5 | 1 | 0 | 2+0 | 1+0 | 0 | 5 | 1 | 0 | 2+0 | 1+0 | 0 |
| | | BF | BF | BF | | | BF | BF | BF | BF | | | BF | BF | BF |

i) if you have any prior information →

✓ start = i

✓ end = i+l-1

[ l — len
  i — s len

ii) if you don't have prior info | window

bruteforce

```
loop( start+1 to end){
    ans = Z[i-start]
    if( ans < remaing window)
        Z[i] = ans
    else if( ans = remaining window)
        Z[i] = [ans + ( find via BF))
    else if( ans > remaining window)
        Z[i] = (remaining window + BF)
}
```

$$\boxed{\begin{array}{l} TC \Rightarrow O(N) \\ SC \Rightarrow O(N) \end{array}}$$

Z-Algo

Q2 Given a string S and a pattern P, find if the pattern exists as a substring in S.

S = mynameis pammi

P = mei

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

str => m e i $ m y n a m e i s p a m m i

x 0 0 0 1 0 0 0 3

true

Q2. Given two strs. A & B. Find the no. of times B occurs in A as a substring.

A = a b a b c a b c a a b c

B = a b c

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

str => a b c $ a b a b c a b c a a b c

2 => x 0 0 0 2 0 3 0 0 3 0 0 1 3 0 0

O/P = 3